Shantanu Shastri

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EDUCATION

Carnegie Mellon University — December 2026
Master of Information Systems Management — GPA: 3.85

Manipal Institute of Technology — August 2022
Bachelor of Electronics and Communications Engineering — GPA: 3.99

Relevant Coursework: Deep Learning, Generative AI Lab, LLM Applications, MLOps, System Design

SKILLS

AI Platform & MLOps: Prompting, Evals, Model Integrations, Observability, MLflow, Docker, Prometheus
GenAI & LLMs: RAG, Vector Search, Semantic Routing, Fine-Tuning, Llama-3, GPT-4, OpenAI APIs
Languages & Backend: TypeScript, Node.js, Python, Java, FastAPI, RESTful APIs, Microservices
Infrastructure & Data: Apache Kafka (Data Pipelines), Milvus, Neo4j, PostgreSQL, Azure, CI/CD

EXPERIENCE

BlackRock — January 2022 – August 2026

AI Engineer Internship — June 2026 – August 2026

- **AI Platform Primitives:** Engineered a GenAI multi-agent platform (RockAI ADK) providing shared primitives and APIs for model integrations, handling 2K+ peak RPS to safely scale AI capabilities.
- **Evals & Observability:** Created offline-online evaluation frameworks and release gates tracking data drift, latency, and confidence scores, utilizing observability to safely process 3.5M records.
- **Context Management Pipeline:** Built a fault-tolerant vector-graph data pipeline (Milvus, Neo4j, Kafka) for real-time context management, enabling sub-100ms structured search and improving latency.

Software Developer 2 - Associate — January 2025 – July 2025

- **LLM Fine-tuning & Integration:** Engineered a TypeScript micro-frontend translating natural language into JQL, fine-tuning LLMs (LoRA) on a 40K+ corpus to achieve a 94% Exact Match quality measurement.
- **Retrieval-Augmented Generation:** Implemented RAG using Milvus vector stores and Cohere semantic reranking to optimize context management, cutting query formulation time by 91%.
- **System Debugging & Latency:** Engineered a distributed read-through cache with LRU eviction to debug and address high-latency APIs, cutting response times from 450ms to 150ms with minimal disruption.

Software Developer 1 - Analyst — July 2022 – December 2024

- **MLOps & Serving Infrastructure:** Built scalable MLOps and serving infrastructure using FastAPI and Docker for predictive modeling systems, achieving 88% precision and reducing manual intervention by 60%.
- **Data Processing Pipelines:** Orchestrated large-scale data processing pipelines to clean multi-terabyte transactional datasets, uncovering latent patterns that optimized downstream system efficiency.
- **Reliability Gates & CI/CD:** Built comprehensive Azure DevOps pipelines with integrated quality gates and JUnit testing suites, ensuring high reliability and reducing deployment time-to-market by 40%.

PROJECTS

Zulip Open-Source GenAI Extensions

- Designed an LLM-backed Topic Drift Detection system to correct conversation titles in real-time, heavily optimizing cost/performance tradeoffs and minimizing API token usage.
- Built a high-throughput message engine using Node.js and Redis to process live data streams into concise summaries, utilizing clear quality and reliability gates to prevent message loss.

Production Recommendation & MLOps System

- Developed a robust MLOps platform featuring a 4-hour retraining loop, zero-downtime model serving, and extensive operational observability via Prometheus and Grafana dashboards.